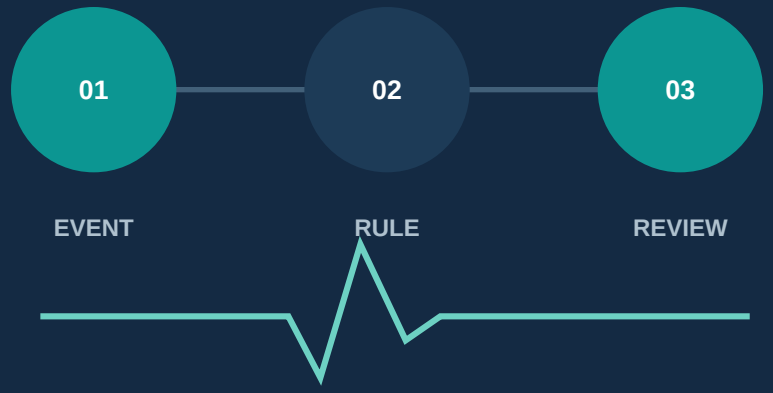
TriagePulse AI

Every second matters. Every signal counts.

A LinuxONE-hosted prototype that turns simulated patient observations into explainable alerts for human review.



IBM LINUXONE

RHEL 9.8 / s390x / SQLite

Synthetic data only | Educational prototype | Not clinically validated

01 / THE PROBLEM

Important changes can hide in a stream of readings.

Emergency teams receive observations over time while patient status and available information can change.

1

OBSERVATION ARRIVES

A new set of measurements is recorded.

2

A SIGNAL CHANGES

Values or patterns may cross a review rule.

3

A PERSON REVIEWS

The alert gives staff a reason to look closer.

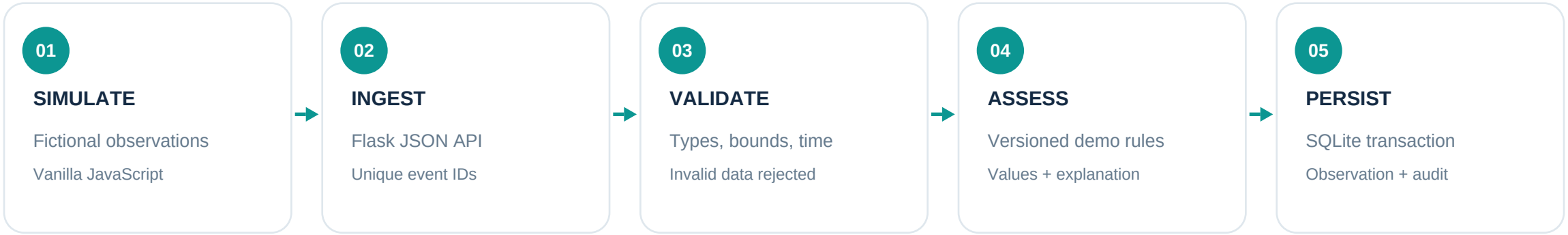
THE PROTOTYPE RESPONSE

Process each event, apply visible rules, preserve the evidence, and make the alert explanation available for human review.

Design principle A rule alert supports attention; it does not make a diagnosis or triage decision.

One event in. An explainable review signal out.

The same API powers the simulator and dashboard; observations, assessments, and alert history persist on the VM.



DEPLOYMENT	SERVICE	ACCESS	DASHBOARD
LinuxONE / RHEL 9.8 / s390x	systemd + Gunicorn	SSH tunnel / loopback only	HTML / CSS / JavaScript

No outcome-labelled dataset is configured; no predictive model is deployed.

Show the rule and the measurement behind it.

Three deterministic simulator paths exercise normal, concerning, and invalid observations through the backend.

A / NORMAL

HR 78

RR 16

SpO2 98%

SBP 120

Temp 36.8 C

No configured rule fires.

B / CONCERNING

HR 145

RR 32

SpO2 88%

SBP 82

Temp 38.1 C

Low SpO2 and SBP rules create an explained alert.

C / INVALID

HR "bad"

Missing fields

Out of order

Duplicate ID

Impossible value

Validation rejects invalid input; it is not treated as a normal reading.

Alert acknowledgement is recorded in audit history and leaves the original observation and assessment intact.

Categories and thresholds are demonstration settings, not approved clinical triage classifications.

Running on IBM Z. Verified end to end.

The service was installed on the target s390x VM, exercised through its API, and checked after a controlled restart.

HOST AND SERVICE

PLATFORM	IBM LinuxONE / RHEL 9.8
ARCHITECTURE	s390x / Python 3.9.25
RUNTIME	Gunicorn 23.0.0 / systemd
DATA	SQLite on persistent VM storage
NETWORK	127.0.0.1:8000 via SSH tunnel

SERIAL INGESTION SAMPLE

100 synthetic POST events

one client / same-host loopback / no concurrency

Sum of measured request times: 0.26 s

Validation 4 API tests passed | persistence, deduplication, alert, and acknowledgement verified

Small smoke-load sample, not a capacity benchmark or performance guarantee.

A working foundation for responsible evaluation.

The prototype demonstrates real-time event handling, persistent explainable alerts, and an IBM LinuxONE deployment path.

WHO COULD BENEFIT

Clinical operations teams can explore how a shared monitoring view might surface observations for review. Datathon teams can inspect a complete event-to-alert workflow on LinuxONE.

WHAT COMES NEXT

- Review demonstration rules with clinical experts.
- Use only authorized, appropriately governed data.
- Define defensible outcomes before training a model.
- Measure concurrency and recovery on target hardware.

PROTOTYPE BOUNDARY

Fictional data only. No clinical validation, diagnosis, treatment recommendation, or autonomous triage. Human professionals remain responsible for care decisions.